

Lab Report-02

Group-01

17B

Md. Tahsin Hossain (24301099)

Zariful Anower Chowdhury (24101334)

S.M. Muhiul Rahman Samir (24321340)

Shabab Hossain Siam (22201939)

Lab Experiment 4

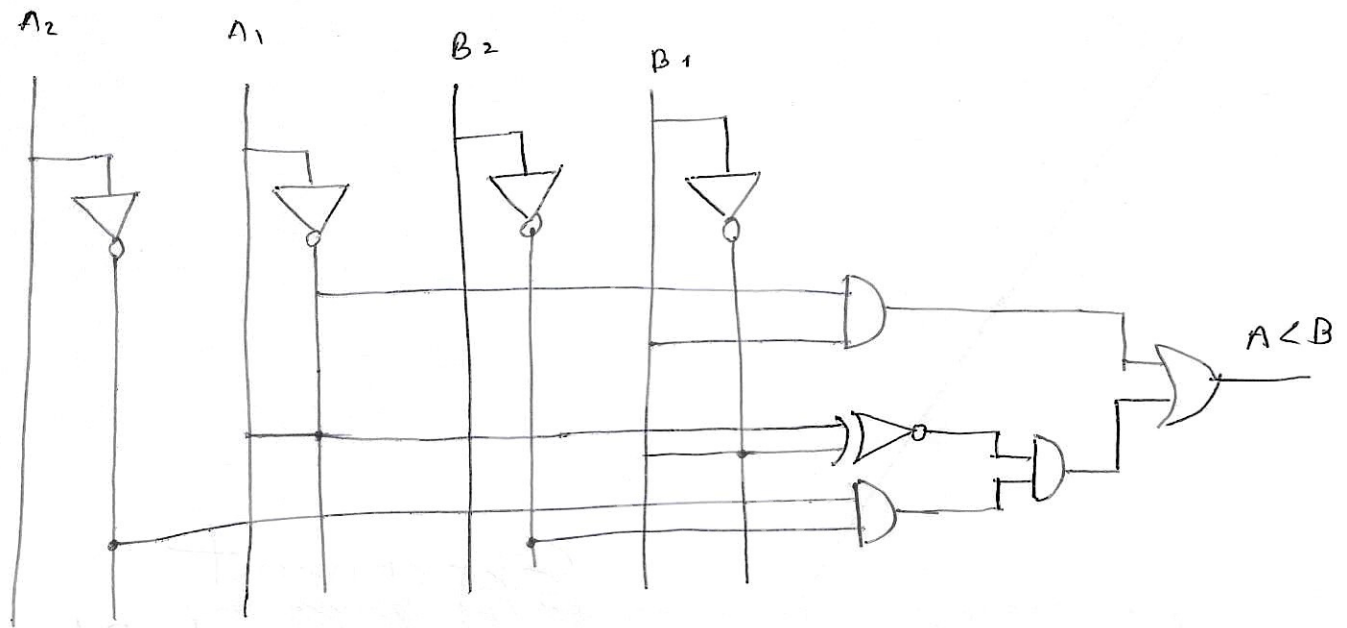
Experiment ^{name} 1: Implementation of 4-bit Magnitude Comparator

Required components:

1. IC 7408
2. IC 7432
3. IC 7404
4. IC 4077

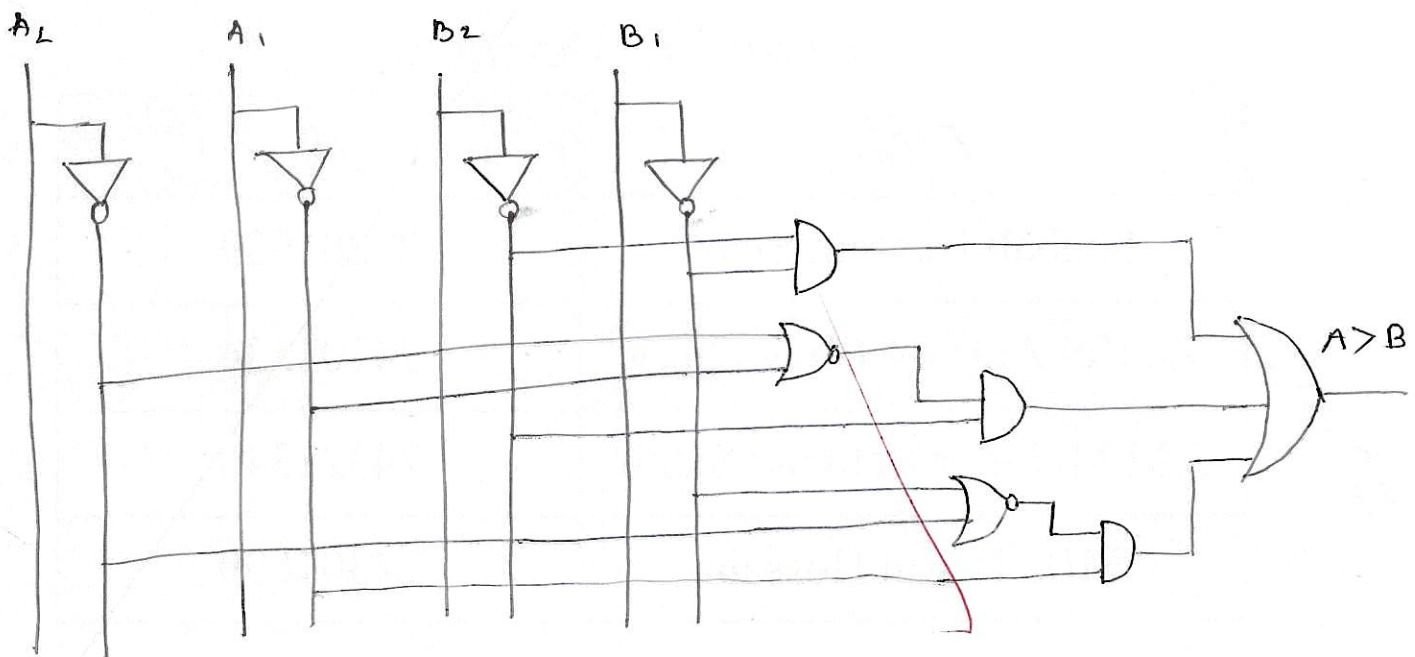
Experimental Setup:

$A < B$ (2 bit)



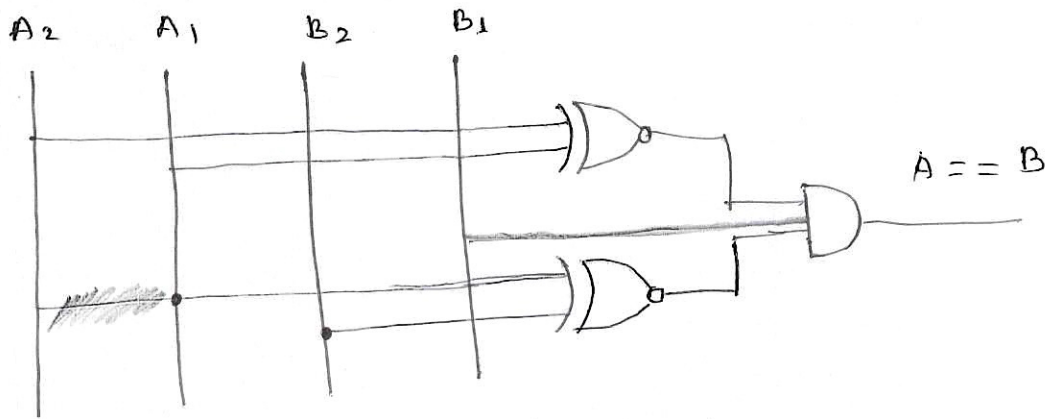
$$A_2' B_2 + (A_2 \odot B_2) A_1' B_1$$

$A > B$ (2 bit)



$$A_2 B_2' + (A_2 \odot B_2) A_1 B_1'$$

$$A == B \text{ (2 bit)}$$



$$(A_2 \odot B_2) (A_1 \odot B_1)$$

Truth Table (2 bit)

	A	B	$A == B$	$A > B$	$A < B$
a.	11	01	0	1	0
b.	00	01	0	0	1
c.	00	00	1	0	0
d.	01	10	0	0	1
e.	01	01	1	0	0
f.	10	01	0	1	0

Handwritten signature in red ink.

Lab Experiment - 5

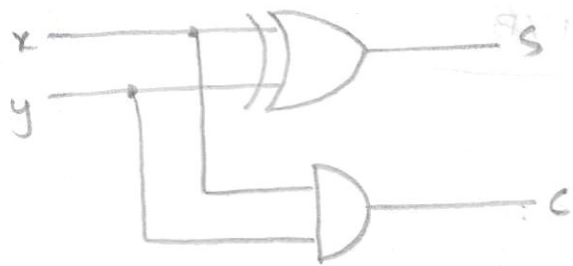
Experiment name: Design & Implementation of 4-bit Parallel Binary Adder.

Required components:

1. IC 7408
2. IC 7432
3. IC 7486
4. IC 7483

Experimental Setup:

Half Adder:

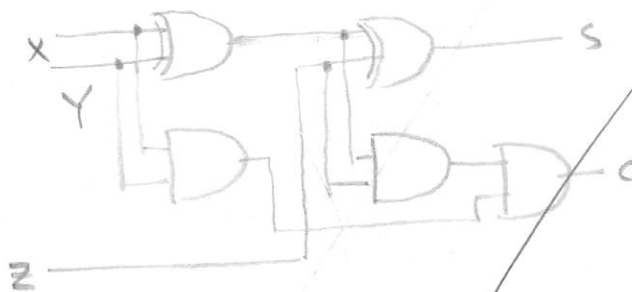


x	y	c	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$S = x \oplus y$$

$$C = xy$$

Full Adder:



x	y	z	c	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$C = xy + (x \oplus y)z$$

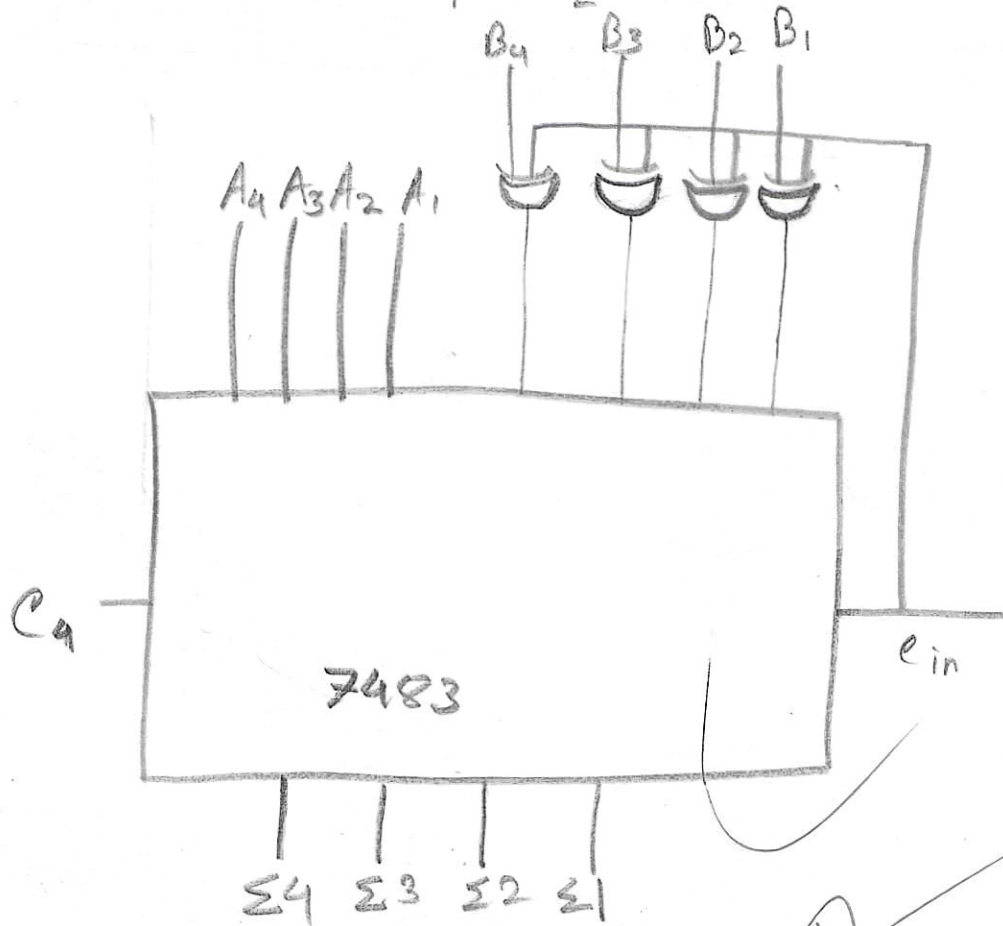
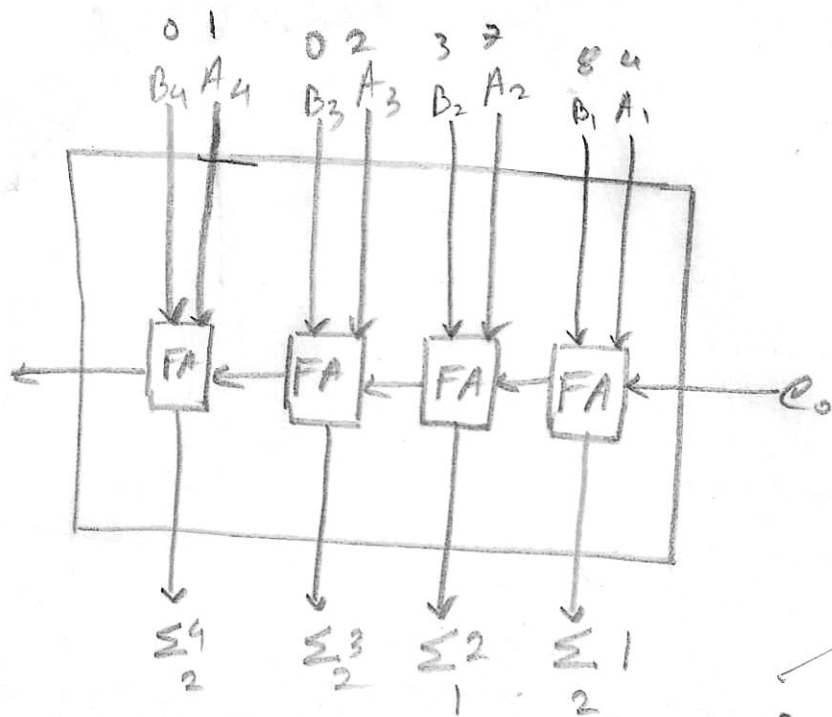
$$S = (x \oplus y) \oplus z$$

4-bit Parallel Adder

	A	B	cin	c4	s4	s3	s2	s1
a.	0000	1000	0	0	1	0	0	0
b.	1100	1101	0	1	1	0	0	1
c.	1110	0011	0	1	0	0	0	1
d.	1111	1111	0	1	1	1	1	0

4-bit Parallel Adder cum Subtractor

	A	B	cin	c4	s4	s3	s2	s1
a.	1100	1000	1	1	0	1	0	0
b.	1100	1101	0	1	1	0	0	1
c.	1110	0011	1	1	1	0	1	1
d.	1111	1111	0	1	1	1	1	0



Handwritten signature

Lab Experiment - 6

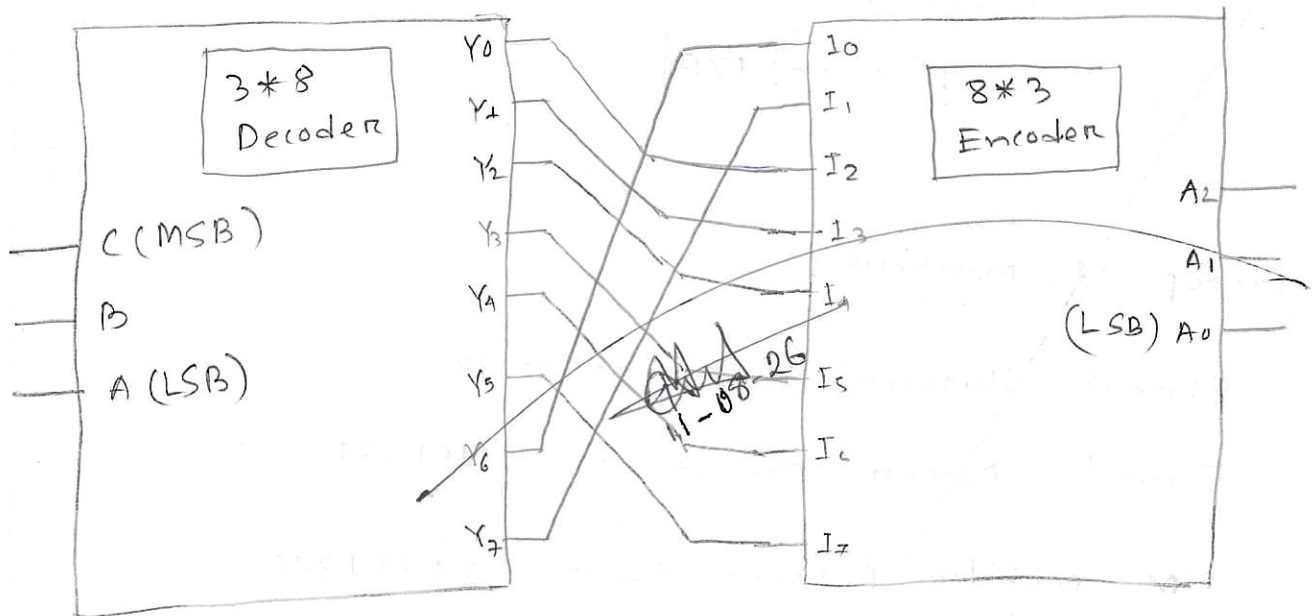
Experiment name: Design circuit using encoder & decoder.

Required components:

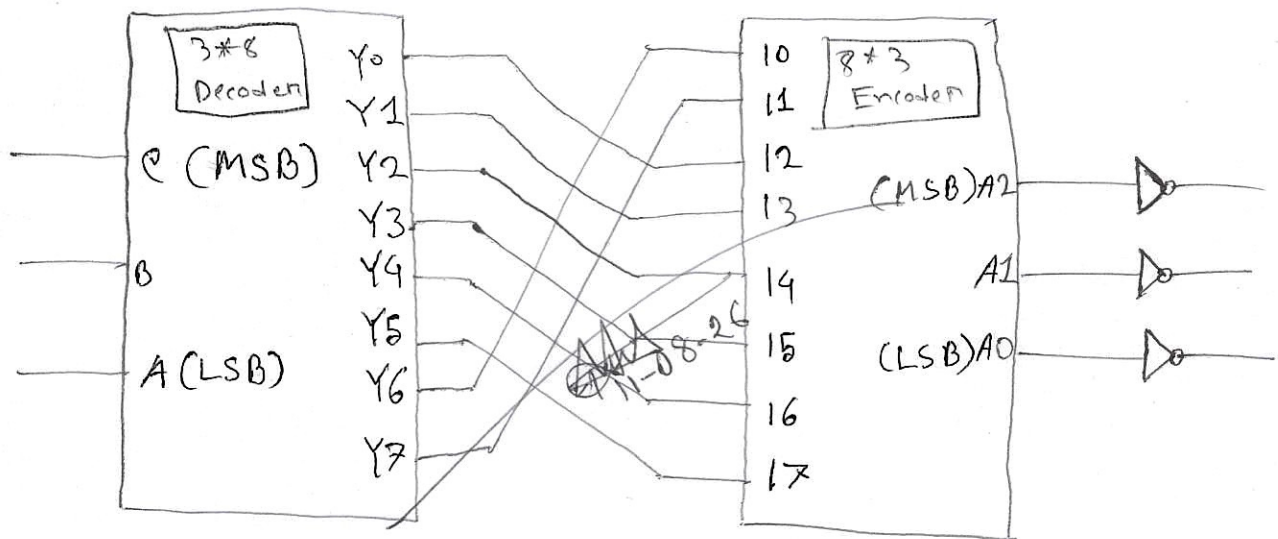
1. IC 74138
2. IC 74148
3. IC 7404

Experimental Setup:

Checkpoint 1 :



Checkpoint-2



Lab Experiment-7

Name of experiment: Function Implementation using
4x1 MUX

Required components:

1. IC 74153
2. IC 7408
3. IC 7432
4. IC 7404

Experimental Setup:

Table For Variable A and B:

	I_0	I_1	I_2	I_3
$A'B'$	0	(1)	(2)	(3)
$A'B$	(4)	(5)	(6)	(7)
AB'	8	(9)	10	(11)
AB	(12)	(13)	14	15

Equations:

$$I_0 = B$$

$$I_2 = A'$$

$$I_1 = 1$$

$$I_3 = \bar{A}\bar{B} + \bar{A}B + A\bar{B}$$

$$= \bar{A} + A\bar{B}$$

$$= (\bar{A} + A)(\bar{A} + B)$$

$$= \bar{A} + \bar{B}$$

~~15-8-26~~

Ans. to the ques. no-01

Input					Output				
Minterms	A	B	C	D	Minterms	A	B	C	D
9	1	0	0	1	5	0	1	0	1
10	1	0	1	0	6	0	1	1	0
11	1	0	1	1	7	0	1	1	1
12	1	1	0	0	8	1	0	0	0
13	1	1	0	1	9	1	0	0	1
14	1	1	1	0	10	1	0	1	0
15	1	1	1	1	11	1	0	1	1

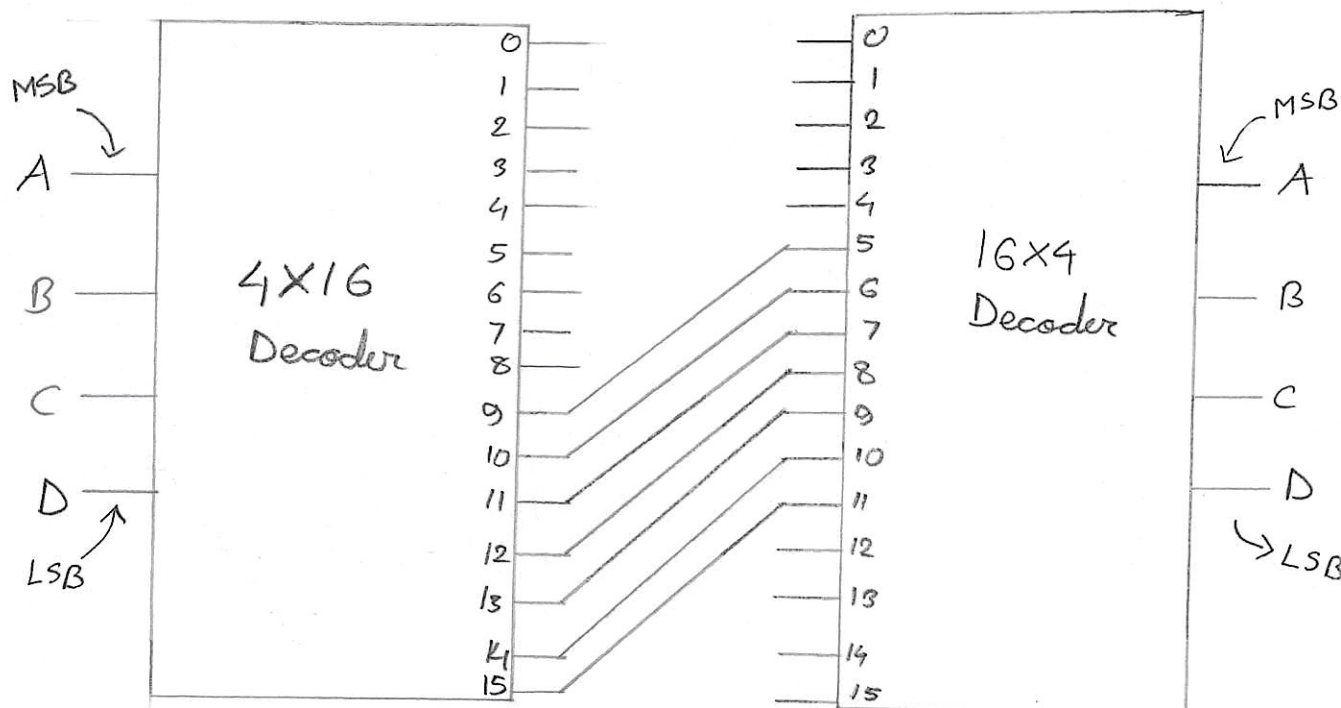


Fig: Excess-9 to Excess-5 converter

Ans. to the ques. no - 02

C_0	B_0	A_0	C_1	S_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$\therefore S_0 = \Sigma (1, 2, 4, 7)$$

$$C_1 = \Sigma (3, 5, 6, 7)$$

Similarly,

$$\therefore S_1 = \Sigma (1, 2, 4, 7)$$

$$C_2 = \Sigma (3, 5, 6, 7)$$

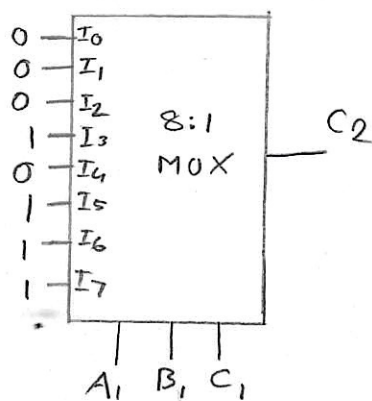
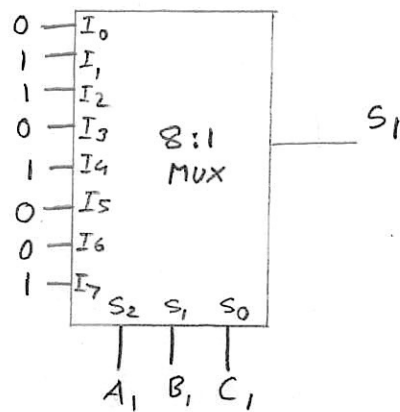
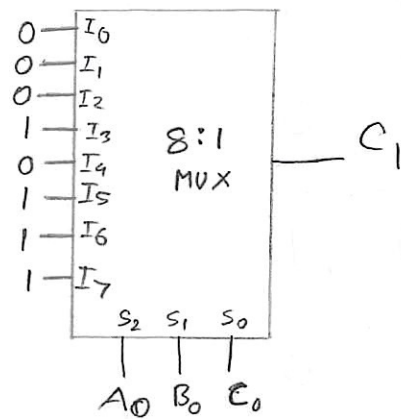
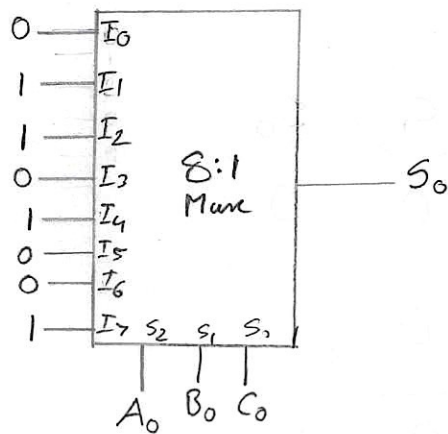


Fig: 2 bit Parallel Adder with 4, 8:1 MUX

Ans. to the ques. no - 03

